

Fermion Mass as Retrodiction Cost: Σ on $J_3(\mathbb{O})$ and the Electroweak Angle

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We propose that fermion mass is the cost of retrodiction failure in the internal (gauge) space of the Standard Model. In the $\Sigma = 2 \ln Q$ framework, the temporal asymmetry $\tau_f = 1 - F_f$ of fermion f in the internal spectral triple (A_F, H_F, D_F) determines its mass via $m_f = M_{\text{ref}} \tau_f$. We identify the internal quantum relative entropy $\Sigma_f = D(\rho_f \| \rho_0)$ with the eigenvalue spectrum of the exceptional Jordan algebra $J_3(\mathbb{O})$, connecting to Singh's parameter-free mass ratios [CONJECTURED]. A key observation: Singh's universal parameter $\delta^2 = 3/8$ for $J_3(\mathbb{O})$ eigenvalue spreading *numerically equals* the Weinberg angle $\sin^2 \theta_W = 3/8$ at GUT scale in the Chamseddine–Connes noncommutative geometry [CONJECTURED]. If these are algebraically identical—both arising from the maximal associative subalgebra $A_F \subset J_3(\mathbb{O})$ (Boyle 2020)—then fermion mass ratios and the electroweak mixing angle share a common information-theoretic origin. We outline the derivation program and state clearly which steps are proven, conjectured, or speculative.

MOTIVATION: RETRODICTION COST IN INTERNAL SPACE

In companion papers [1–3], we established that the quantum relative entropy (QRE)

$$\Sigma = D(\rho_{\text{st}} \| \rho_{\text{mat}}) \geq 0 \quad (1)$$

governs the arrow of time ($\tau = 1 - F$, Paper I [PROVEN]), the gravitational metric ($\Sigma_{\text{grav}} = 2 \ln Q$, Paper II [PROVEN]), and the dimensionality of spacetime ($d = 4$ from the spectral Fisher metric, Paper 9 [PROVEN]).

The Standard Model lives on the almost-commutative geometry $M \times F$ of Connes [4], where F is the finite internal space with algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$. The finite Dirac operator D_F encodes Yukawa couplings, hence fermion masses [KNOWN].

We now extend Σ to the internal space:

$$\Sigma_f = D(\rho_f \| \rho_0)_F, \quad (2)$$

where ρ_f is the state of fermion f in H_F , $\rho_0 = I / \dim(H_F)$ is the vacuum reference, and the subscript F denotes restriction to the internal algebra [CONJECTURED].

The fermion mass is then the retrodiction cost:

$$\boxed{m_f = M_{\text{ref}} \times \tau_f = M_{\text{ref}} \times (1 - e^{-\Sigma_f/2})} \quad (3)$$

with M_{ref} a reference scale identified with the electroweak VEV $v = 246 \text{ GeV}$ [CONJECTURED].

Eq. (3) has the correct limiting behavior:

- $\Sigma_f = 0$: massless (perfect retrodiction, $F_f = 1$, e.g., unbroken gauge symmetry);
- $\Sigma_f \gg 1$: $m_f \rightarrow M_{\text{ref}}$ (saturation; the top quark at $m_t/v \approx 0.70$ is near this limit);
- $\Sigma_f \ll 1$: $m_f \approx M_{\text{ref}} \Sigma_f/2$ (linear regime for light fermions).

Σ_f FROM THE SPECTRAL ACTION

Chamseddine, Connes, and van Suijlekom [5] proved that the von Neumann entropy of the fermionic second quantization of a spectral triple equals a spectral action: $S_{\text{vN}}(\rho_\beta) = \text{Tr}(h(\beta D))$ [KNOWN].

The spectral action on F expands as

$$S_F = \text{Tr}(f(D_F^2/\Lambda^2)) \approx f_0 \text{Tr}(1) + f_2 \Lambda^{-2} \text{Tr}(D_F^2) + \dots \quad (4)$$

The leading mass-dependent term is [KNOWN]

$$\text{Tr}(D_F^2) = \sum_f N_f m_f^2 = v^2 \sum_f N_f y_f^2, \quad (5)$$

where N_f counts color and chirality multiplicities and y_f is the Yukawa coupling.

Identifying Σ_f with the contribution of fermion f to the QRE of the internal state:

$$\Sigma_f \propto y_f^2 = (m_f/v)^2 \quad [\text{CONJECTURED}] \quad (6)$$

This is consistent with Eq. (3) in the light-fermion regime ($\Sigma_f \ll 1$): $m_f \approx v \Sigma_f/2 \propto v y_f^2/2$, which gives $m_f \propto y_f^2$ —differing from the usual $m_f = y_f v/\sqrt{2}$. The resolution is that Eq. (3) determines τ_f (the dimensionless asymmetry), not m_f directly. The correct identification is:

$$\tau_f = y_f/\sqrt{2} \implies \Sigma_f = -2 \ln(1 - y_f/\sqrt{2}). \quad (7)$$

For the top quark, $y_t/\sqrt{2} \approx 0.70$, giving $\Sigma_t \approx 2.41$ —a large but finite retrodiction cost [CONJECTURED].

CONNECTION TO $J_3(\mathbb{O})$ MASS RATIOS

Singh [6] showed that the complexified exceptional Jordan algebra $J_3(\mathbb{O}_{\mathbb{C}})$ provides parameter-free fermion

mass ratios with a single universal parameter $\delta^2 = 3/8$. Three generations arise as three eigenvalues of a Jordan element $X \in J_3(\mathbb{O})$; the mass hierarchy comes from a “minimal universal ladder” in $\text{Sym}^3(3)$ of the flavor $\text{SU}(3)$ [KNOWN].

We observe that $\delta^2 = 3/8$ *numerically equals* the Weinberg angle at GUT scale in the Chamseddine–Connes framework [4]:

$$\sin^2 \theta_W|_{\text{GUT}} = \frac{\text{Tr}(T_3^2)}{\text{Tr}(T_3^2) + \text{Tr}(Y^2)} = \frac{3}{8}. \quad (8)$$

Both arise from the algebraic structure of the Standard Model:

- $\sin^2 \theta_W = 3/8$: from the fermion hypercharge assignments dictated by $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ [KNOWN].
- $\delta^2 = 3/8$: from the $\text{Sym}^3(3)$ structure in $J_3(\mathbb{O}_{\mathbb{C}})$ [KNOWN, Singh 2025].

Boyle [7] showed that A_F is the maximal associative subalgebra of $J_3(\mathbb{O})$. This establishes that both $3/8$ ’s live in the *same algebraic structure* [KNOWN].

Conjecture. *The two instances of $3/8$ are algebraically identical: $\delta^2 = \sin^2 \theta_W$ because both express the ratio of abelian to total gauge content within $A_F \subset J_3(\mathbb{O})$. [CONJECTURED]*

If true, the electroweak mixing angle and the fermion mass hierarchy share a common origin in the exceptional Jordan algebra—unified by the retrodiction cost Σ on the internal space.

THE PETZ MAP ON $J_3(\mathbb{O})$

The Petz recovery map $\mathcal{R}$ is the unique retrodiction functor satisfying the detailed balance condition [1, 11]. On $J_3(\mathbb{O})$, we define [SPECULATIVE]:

$$\mathcal{R}_\sigma : J_3(\mathbb{O}) \rightarrow J_3(\mathbb{O}), \quad \mathcal{R}_\sigma(X) = \sigma^{1/2} X \sigma^{1/2}, \quad (9)$$

where σ is a reference state in the Jordan algebra (the product is the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$).

The recovery fidelity for generation i :

$$F_i = F(\rho_i, \mathcal{R}_\sigma(\rho_i)) = e^{-\Sigma_i/2}. \quad (10)$$

Speculative identification. Singh’s “minimal universal ladder” = the Petz recovery channel $\mathcal{R}_\sigma$ on $J_3(\mathbb{O})$ that maximizes the average fidelity $\bar{F} = \frac{1}{3} \sum_{i=1}^3 F_i$ subject to the Jordan algebra constraints.

If this holds, the mass ratios follow from Petz optimality, and the CKM matrix elements are the fidelity overlaps between up-type and down-type Petz channels:

$$|V_{ij}|^2 = F(\mathcal{R}_\sigma^{(u,i)}, \mathcal{R}_\sigma^{(d,j)}) \quad [\text{SPECULATIVE}]. \quad (11)$$

TABLE I. Retrodiction cost Σ_f and temporal asymmetry τ_f for SM fermions, using $M_{\text{ref}} = v = 246$ GeV. Values computed from $\tau_f = m_f/v$ and $\Sigma_f = -2 \ln(1 - \tau_f)$.

Fermion	m_f (GeV)	τ_f	Σ_f
ν (lightest)	$\lesssim 10^{-11}$	$\lesssim 4 \times 10^{-14}$	$\lesssim 10^{-13}$
e	5.11×10^{-4}	2.08×10^{-6}	4.15×10^{-6}
u	2.2×10^{-3}	8.9×10^{-6}	1.79×10^{-5}
d	4.7×10^{-3}	1.9×10^{-5}	3.83×10^{-5}
μ	0.106	4.31×10^{-4}	8.62×10^{-4}
s	0.093	3.78×10^{-4}	7.57×10^{-4}
c	1.27	5.16×10^{-3}	1.03×10^{-2}
τ	1.777	7.22×10^{-3}	1.45×10^{-2}
b	4.18	1.70×10^{-2}	3.43×10^{-2}
t	173	0.703	2.41

MASS TABLE AND PREDICTIONS

Table I lists the retrodiction cost Σ_f for each SM fermion, computed from Eq. (7) with $\tau_f = m_f/v$. The top quark has $\Sigma_t = 2.41$, making it the most “internally irreversible” fermion. The electron has $\Sigma_e = 4.15 \times 10^{-6}$: it is nearly perfectly retrodictable in the internal space. The neutrino has $\Sigma_\nu \lesssim 10^{-13}$: it is the closest to a “zero-entropy” internal state.

Testable prediction. $m_f \leq v$ for all SM fermions (Petz bound saturation with $M_{\text{ref}} = v$). This is satisfied: $m_t = 173 \text{ GeV} < 246 \text{ GeV}$ [CONJECTURED, consistent with data].

Open prediction. If the $\delta^2 = \sin^2 \theta_W$ identification holds, and Singh’s mass ratios are correct, then the Σ framework *with no free parameters* predicts all fermion mass ratios from the algebraic structure of $J_3(\mathbb{O})$.

DISCUSSION AND STATUS

We summarize the logical status of each claim:

PROVEN:: $\tau = 1 - F$, $\Sigma_{\text{grav}} = 2 \ln Q$, $g_{QQ} = 2/Q^2$ in $d = 4$, $\sin^2 \theta_W = 3/8$ in NCG, spectral action = S_{vN} , Singh’s mass ratios from $J_3(\mathbb{O})$.

CONJECTURED:: $\Sigma_f = D(\rho_f \parallel \rho_0)_F$ defines the internal retrodiction cost; $m_f = v \tau_f$; $\delta^2 = \sin^2 \theta_W$ is algebraically the same $3/8$; the Petz bound gives $m_f \leq v$.

SPECULATIVE:: Σ extremization on $J_3(\mathbb{O})$ selects the physical mass ratios; Singh’s ladder = Petz recovery on $J_3(\mathbb{O})$; CKM matrix = Petz fidelity between up/down channels.

The key testable step is the algebraic identification of the two $3/8$ ’s. If the Boyle embedding $A_F \subset J_3(\mathbb{O})$ maps

Singh's δ^2 to the NCG $\sin^2 \theta_W$, this would unify the electroweak mixing angle and the fermion mass hierarchy within a single information-theoretic framework.

Concretely, one must show that the $\text{Sym}^3(3)$ eigenvalue spread parameter of $J_3(\mathbb{O})$ reduces to $\text{Tr}(T_3^2)/[\text{Tr}(T_3^2) + \text{Tr}(Y^2)]$ when restricted to A_F . This is a finite-dimensional algebraic computation that we leave to a subsequent paper.

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